

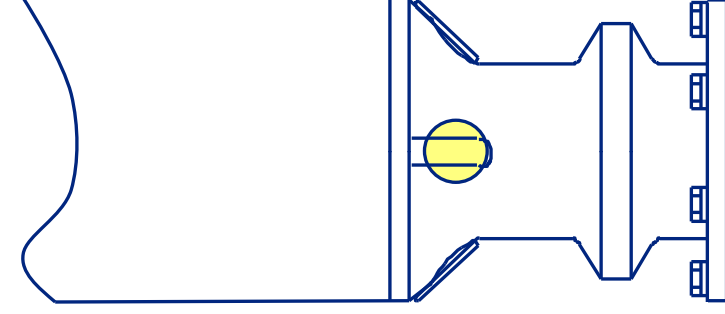
Intakes

There are three types of intakes:

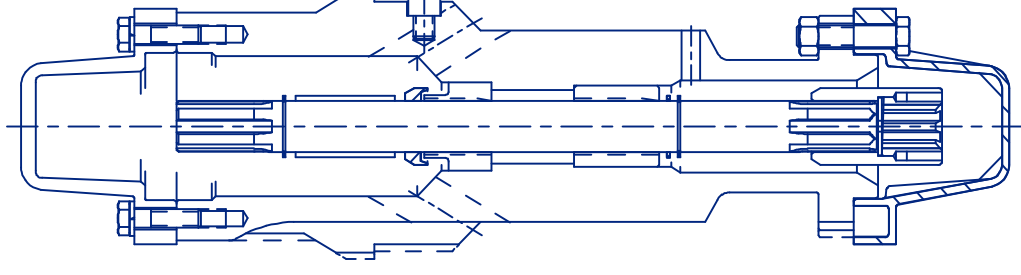
1. Standard Intakes or BOI (bolt on intake)
2. Integral
 - Manufactured as part of the pump
3. Gas Separators

Standard Intake

Standard, BOI and integral intakes do not separate gas. Some gas separation might occur, but it will only be natural separation due to some of the gas not turning and going into the intake when the rest of the fluid does.

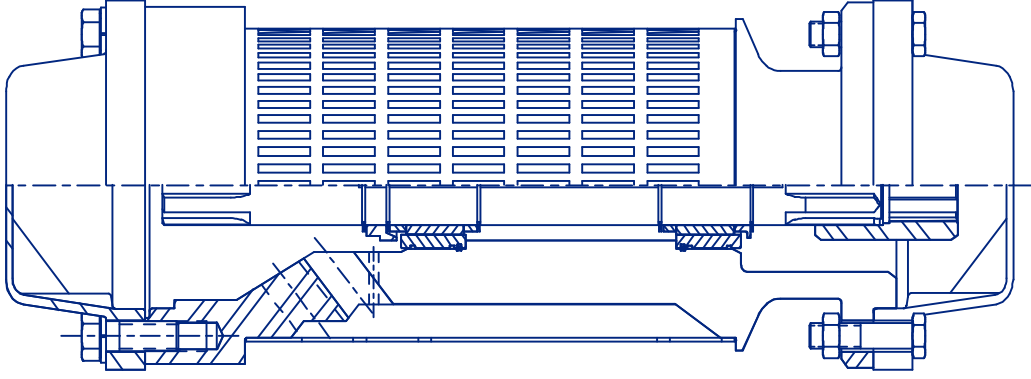


Integral Intake



Bolt-on Intake

Standard Intake



Intakes can be either ARZ, which is the preferred configuration, or standard

The ARZ intake uses Zirconia bearings and sleeves to better protect against abrasive wear and lateral vibration. This is important near a protector seal.

ARZ Intake

Engineering recommends that only ARZ type intakes should be used, because the reliability of the unit (especially the protector) will be greatly enhanced by the durability of the compliant Zirconia bearings.

ARZ Intakes are available in 338, 387, 400, 540, 562, and 675 series



Gas as a Problem

Gas Production has been a “problem” since the early days of oil production. It limited production on many Oil wells producing with pumps.



Historical Solutions

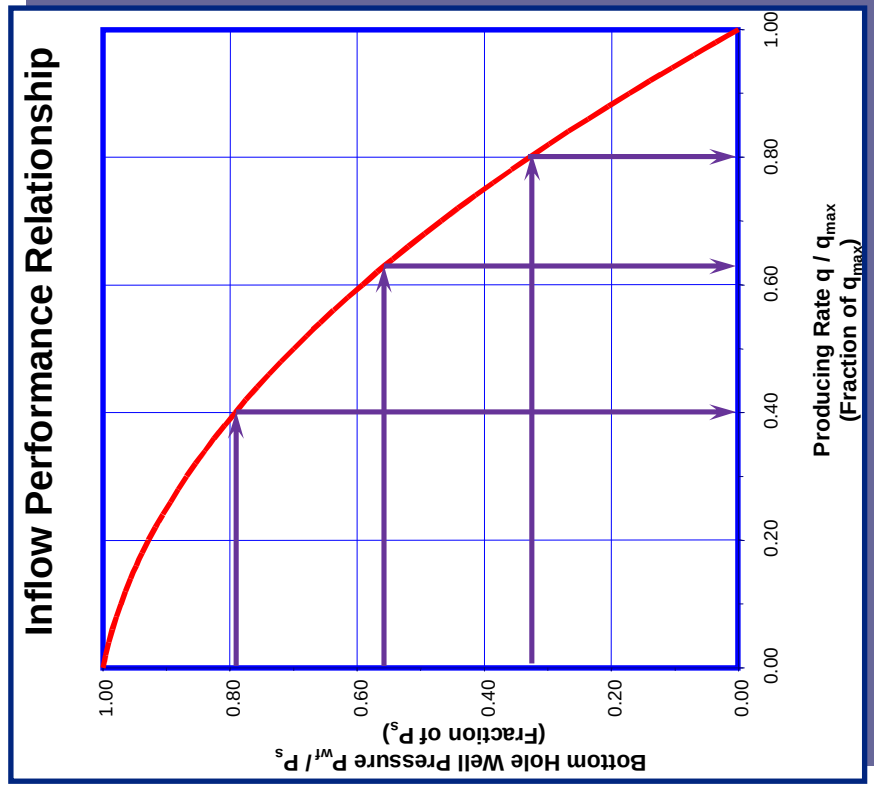
- Gas separators
- Shrouds
- Gas Avoiders
- Tapered pumps
- Advanced Gas Handlers
- Poseidon

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Why is Production Limited?

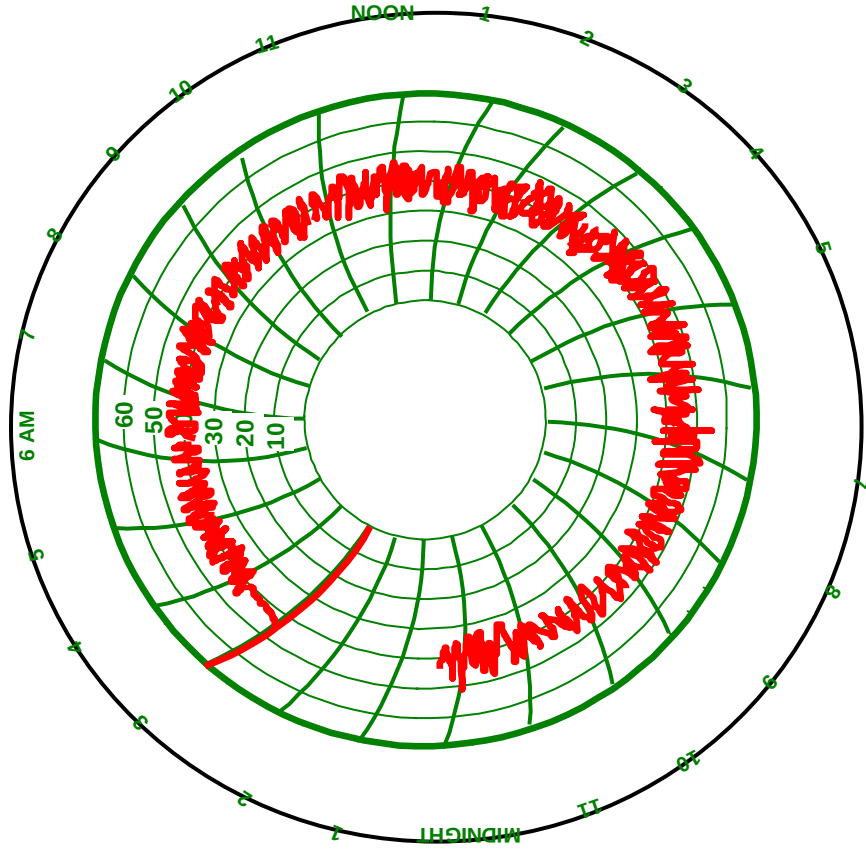
- If the ESP is modified to handle more free gas the amount of draw down can be increased.



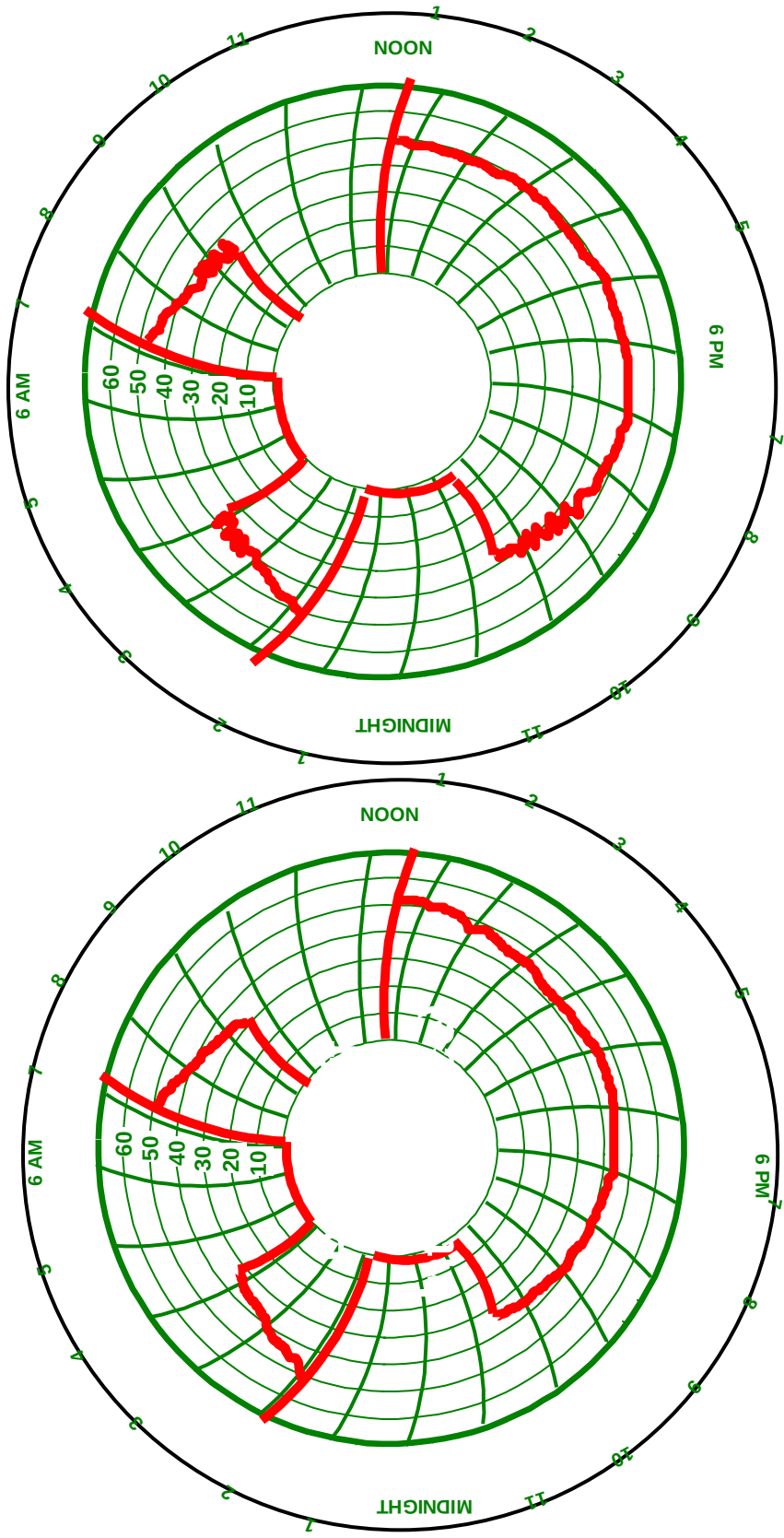
Centrifugal Pump Gas Handling Capacity

Radial Flow	10 %
Mixed Flow	20-25 %
Axial Flow	75 %

Identifying the problem - Amp Charts



Identifying the problem - Amp Charts



Gas Separator

- A gas separator is still an intake, but with some special features designed to keep free gas from entering the pump.
- Two types of Gas Separators
 - Static
 - Dynamic

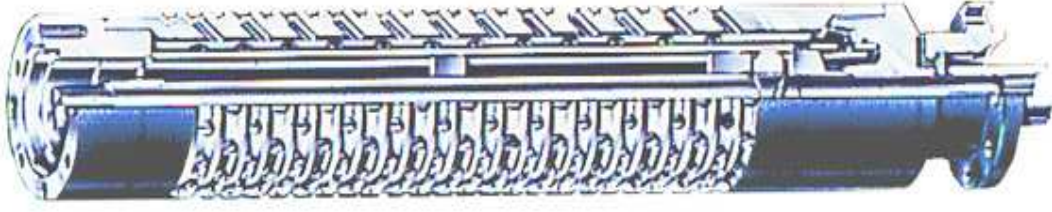
Static Gas Separators

Original gas separator designs were based on increasing gas separation by forcing the fluid flow to reverse in the wellbore. This is where the name of this type of gas separator, REVERSE FLOW, comes from.

Since this type of gas separator does no real "work" on the fluid, it is also called a "static" gas separator.

65 GS

Reverse Flow Gas Separator

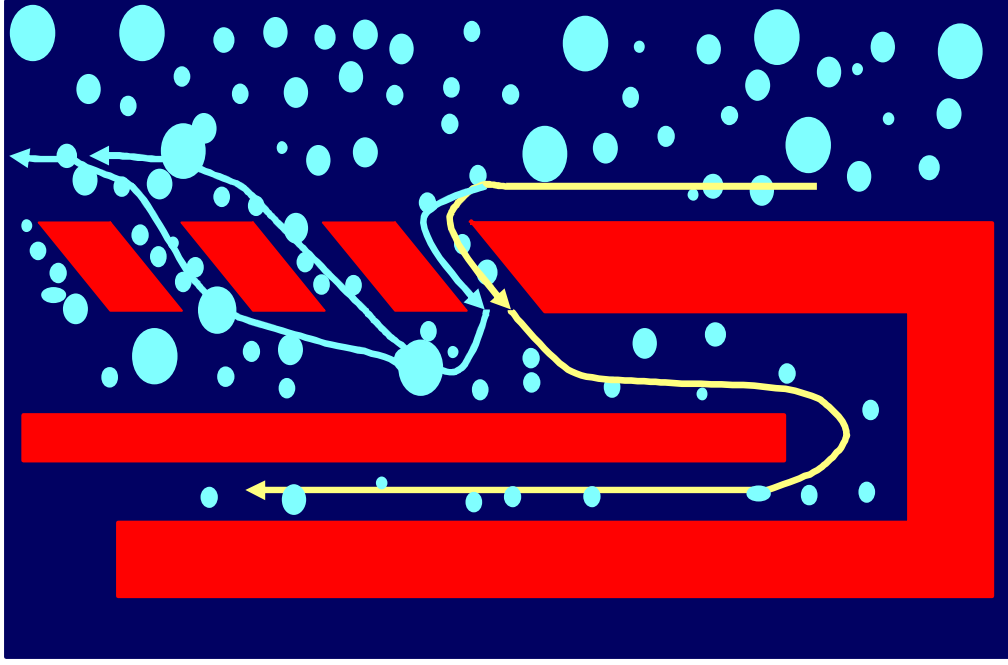


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Static (Reverse Flow) Gas Separator

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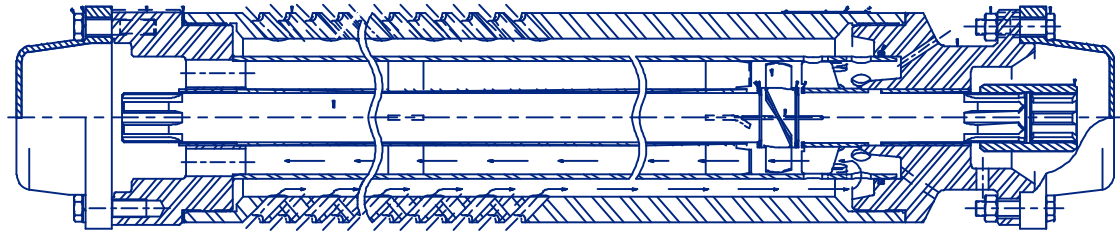


As well fluid enters the gas separator it is forced to change direction. Some of the gas bubbles continue to rise instead of turn or rise inside of the gas separator, exit the housing and continue to rise.

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Reverse Flow Type Gas Separator

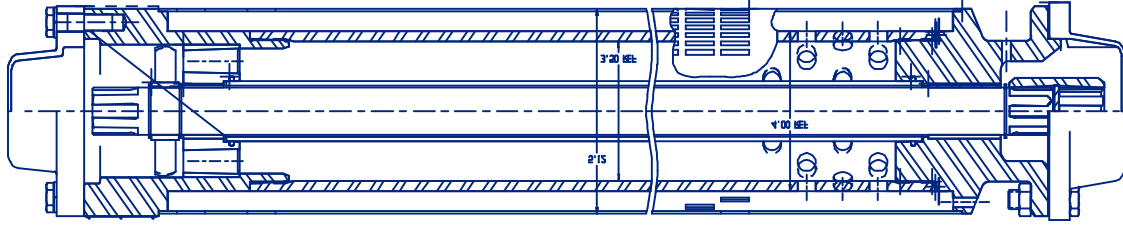
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Angle drilled holes

65 Type

Static Gas Separator



Angle drilled holes

RF Type

Static Gas Separator

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Dynamic Gas Separators

The rotary gas separator design works in a similar fashion to a centrifuge. The first model had centrifuge "paddles" spinning at 3500 rpm and was believed to cause the heavier fluids to be forced to the outside, through the crossover and up into the pump, while the lighter fluid (vapor) stays toward the center, and exits through the crossover and discharge ports back into the well.

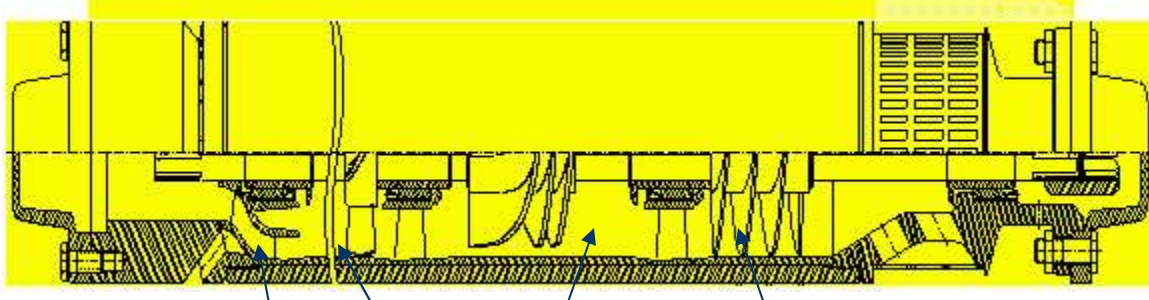


Dynamic Vortex Gas Separator

Discharge Crossover

Multiple vortex generators

Axial flow
inducers



Gas Separator Efficiency

When we talk about gas separator performance we are normally interested first in separation efficiency.

$$\text{Separation Efficiency} = \frac{\text{Free gas separated}}{\text{Free gas available @ suction}}$$

Gas Separators Efficiency

We might also talk about gas separator performance in terms of "gas ingestion percentage"

GIP = 1 - Separation Efficiency.

GIP quantifies how much of the free gas present at the intake would go into the pump.

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Factors Impacting Separation

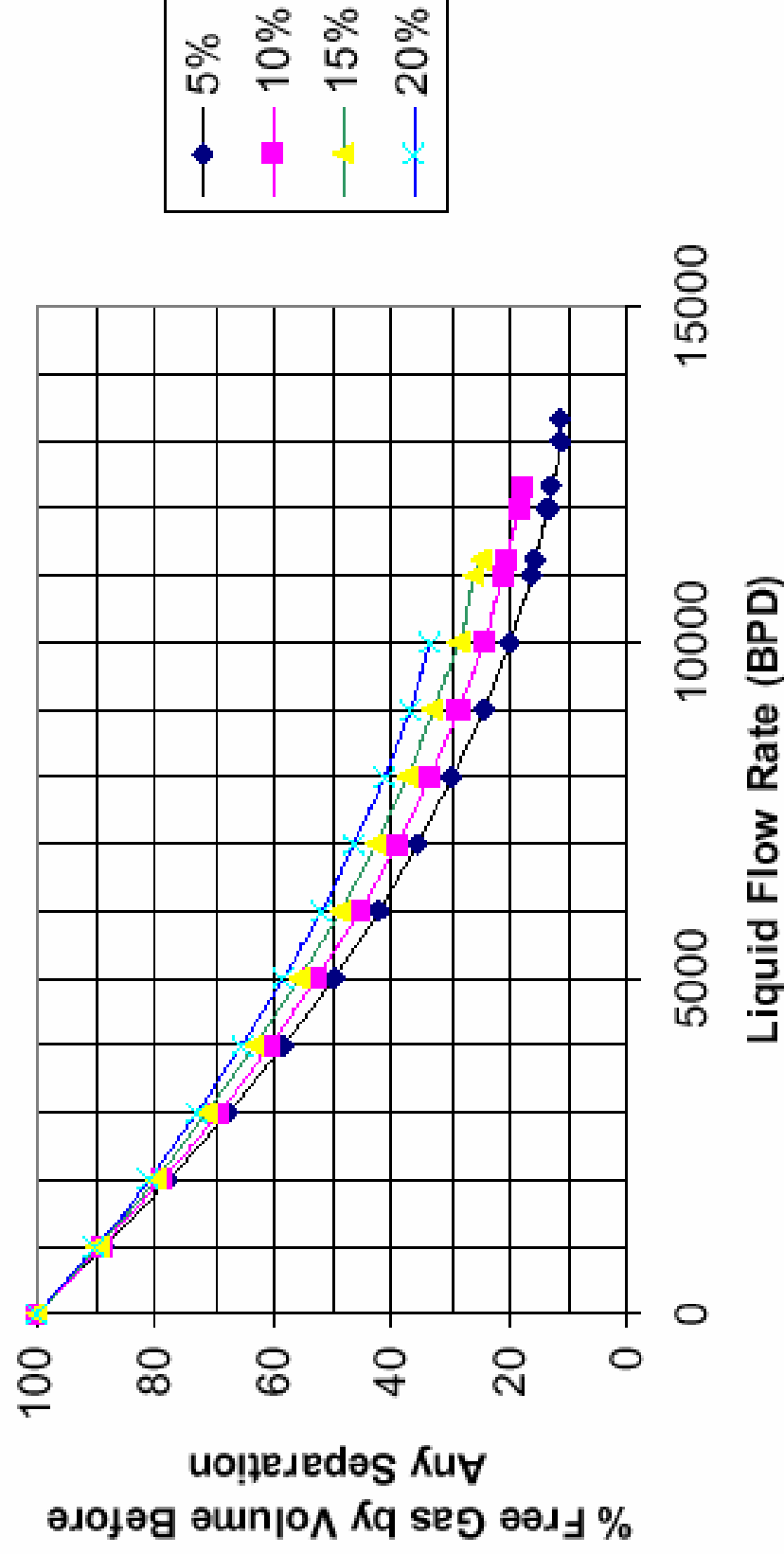
- Hydrocarbon Components
- Casing Size and Equipment size
- % Free Gas by Volume at Intake
- Pump Intake Pressure and Temperature
- GLR
- Bubble Point, etc.
- Flowrate
- Construction of Mechanical Separator
- Equipment Obstructing (Y-Tool)
- Frequency
- Casing Pressure
- Wellbore Orientation

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Performance Curves

VGSA S70-150 Performance Curve for % Free Gas
by Volume After Total Separation



The advanced gas handler (AGH)

The primary aim of the Advanced Gas Handling system is to avoid "gas-locking" in the pump which can result in inefficient cyclical operations and possible premature mechanical failure if not properly controlled.

The AGH does not separate gas but conditions it so it can be produced by the pump.

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The Advanced Gas Handler (AGH)

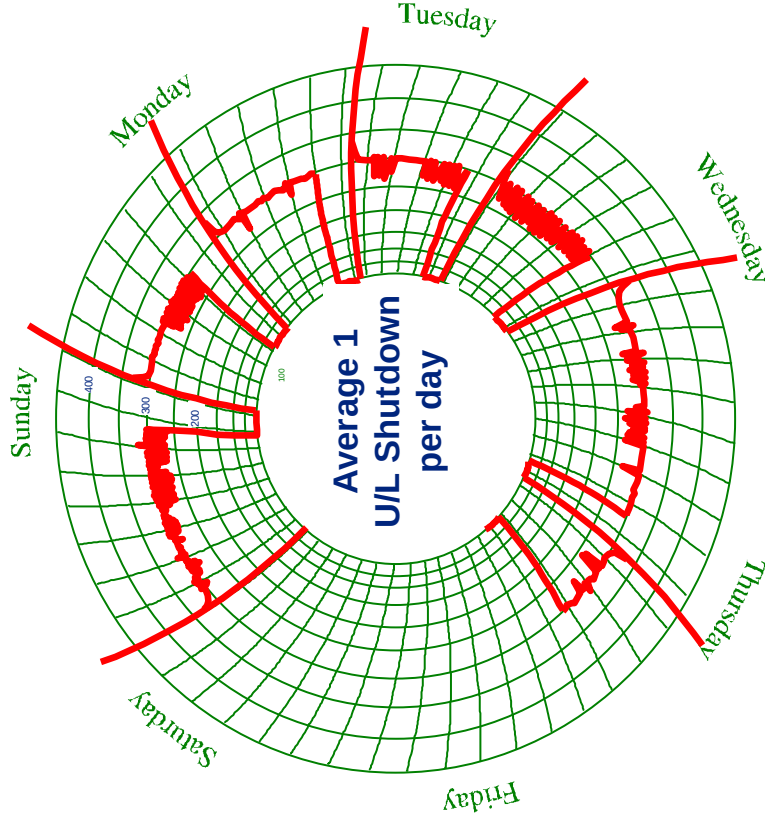
The AGH improves the overall efficiency of many submersible lift systems in comparison to those employing gas separators only. In many cases, the AGH will allow successful production of wells that previously could not be economically or reliably produced with submersible pumps employing gas separators due to gas problems.

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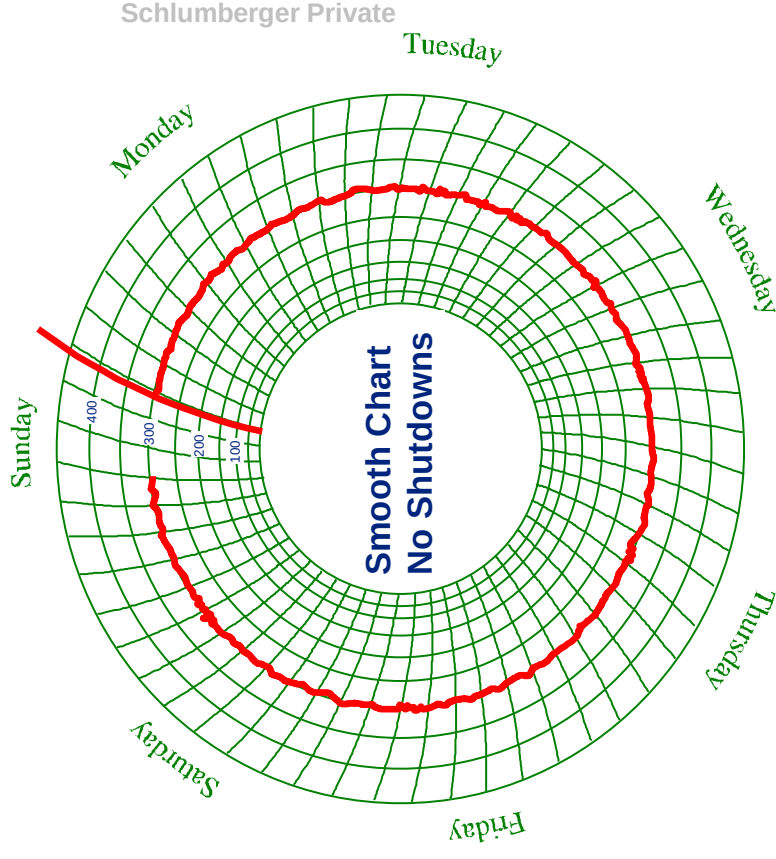
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REDA AGH Performance

The AGH results in stable operation with reduced restarts due to underload shutdown (i.e. gas lock). This improves production and enhances reliability.



With Rotary Separator Only



With AGH and Separator

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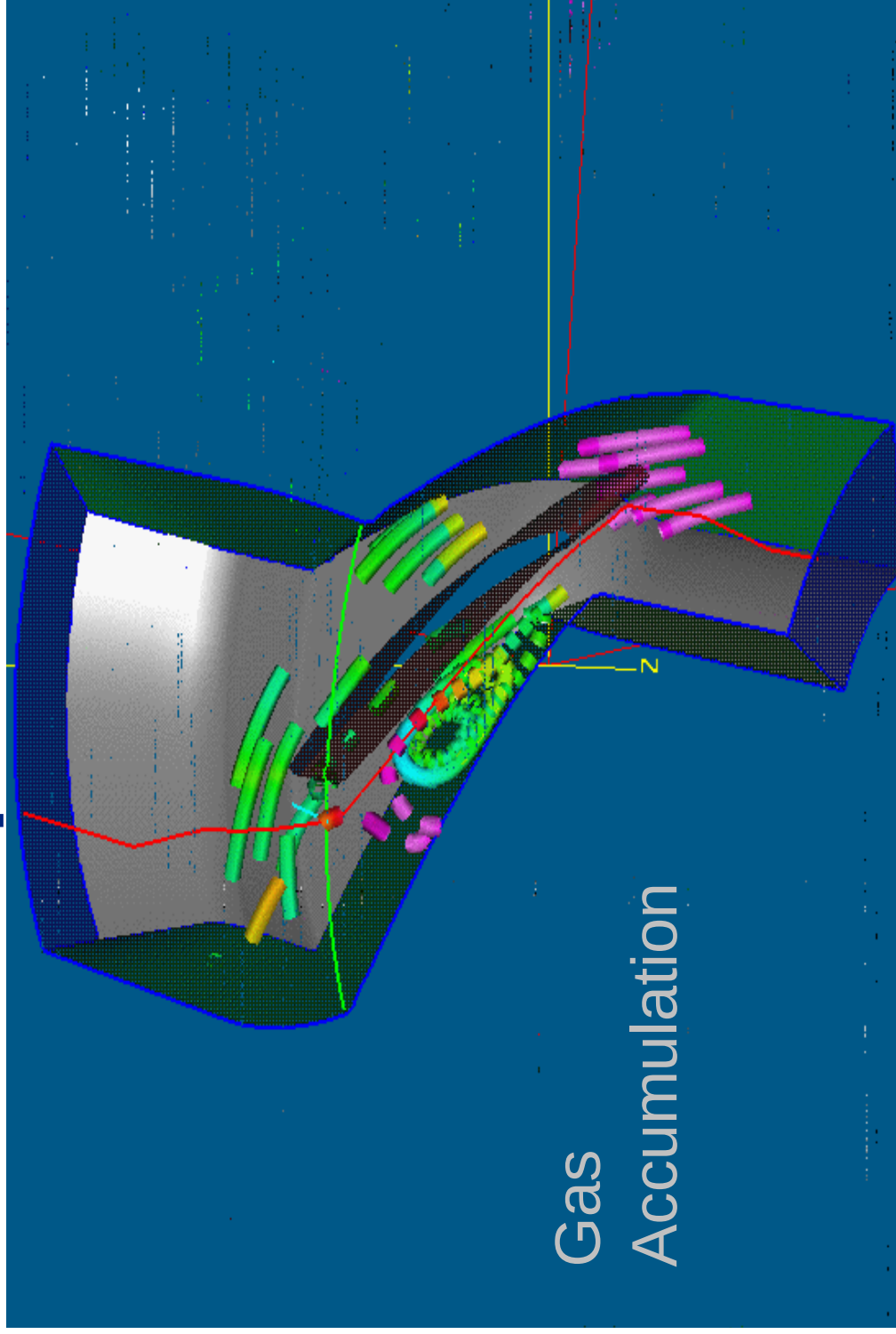
The Advanced Gas Handler (AGH)

The AGH is designed to improve the overall lift efficiency of a submersible lift installation by maintaining a higher gas to liquid ratio in the tubing string. The higher GLR will in act to reduce the hydraulic horsepower required to lift an oil well producing oil, water and gas.

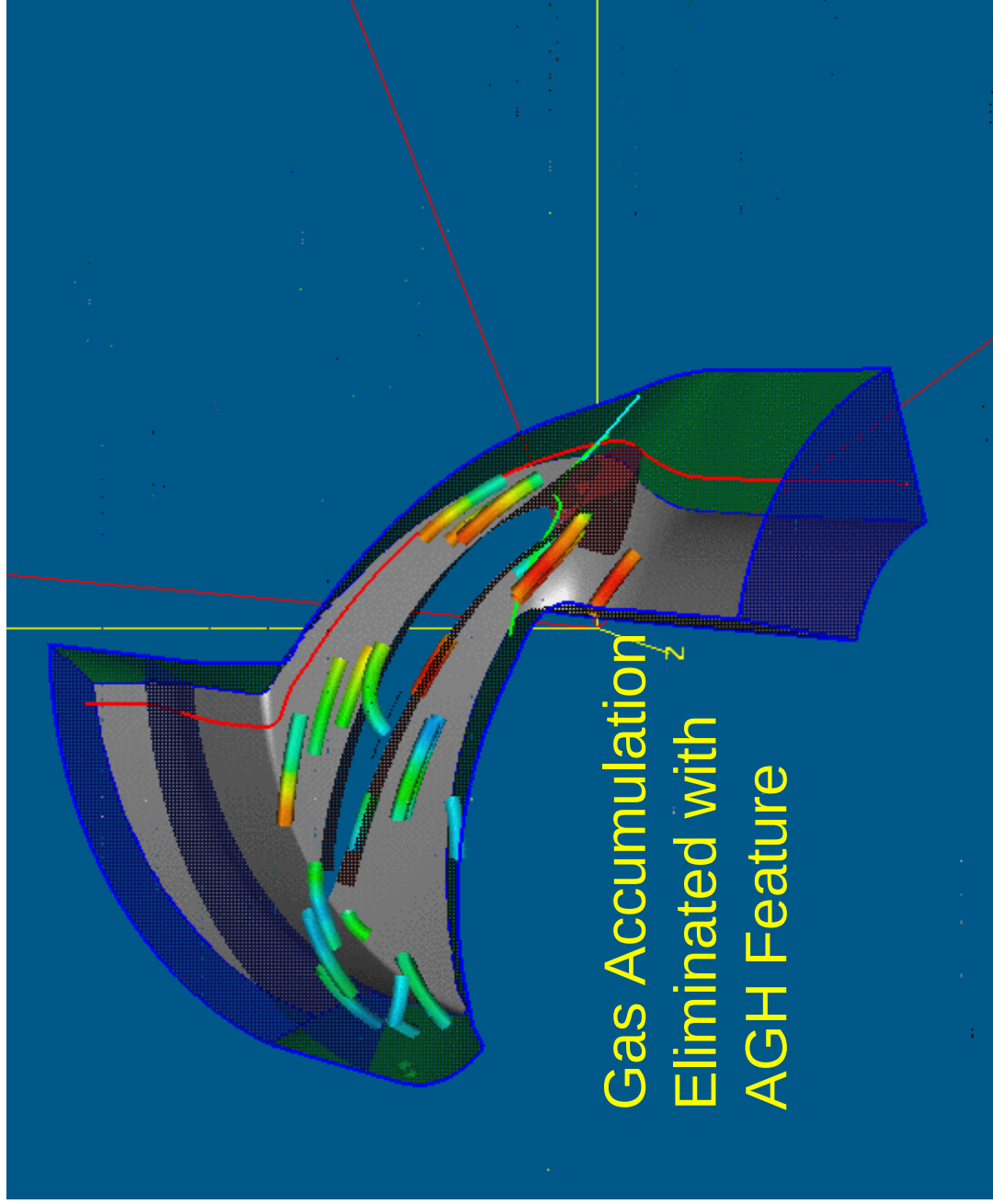
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Standard Impeller



AGH Impeller



AGH Principle

Increase gas handling ability with minimal head loss

Methods:

- Homogenize the mixture
- Reduce bubble size
- Put gas back into solution
- Help gas to move to main stream

Application of AGH

In general, an AGH should be considered if the following conditions exist at the intake of the pump:

Free Gas Percentage = from 30% to 60% by volume

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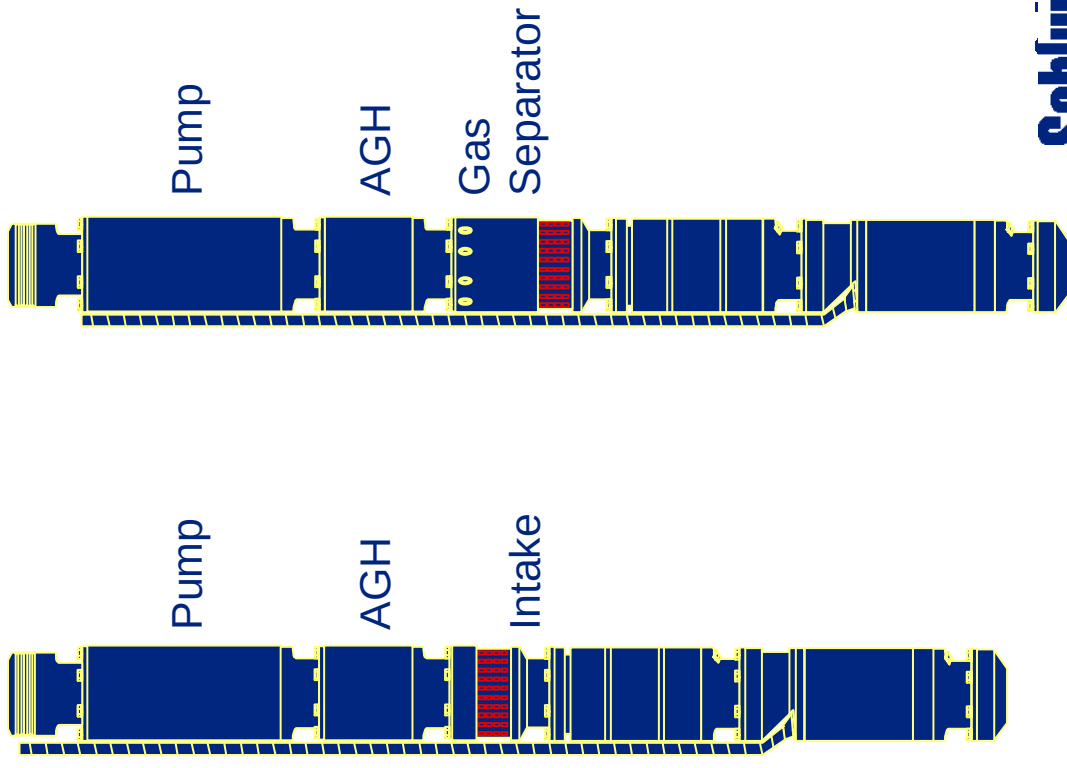
ADVANCED GAS HANDLER DATA

Model	OD	Shaft OD	Shaft Area	Shaft HP	HP	Operating Flowrate
	[inches]	[inches]	[sq.-inch]	limit @ 60Hz.	@ 60Hz.	[bpd]
DN 5-21	4.00	0.6875	0.371	200	13	500 - 2,100
DN 20-60	4.00	0.875	0.601	410	24	2,000 - 6,100
GN 20-40	5.13	1	0.785	600	37.5	2,000 - 4,000
GN 40-80	5.13	1	0.785	600	45	4,000 - 8,000
SN 70-100	5.38	1	0.785	600	53	7,000 - 10,000
HN 100-250	5.62	<u>1.19</u>	1.108	<u>637</u>	102	10,000 - 25,000

Application of AGH

The AGH can be used with a standard intake or with a gas separator.

The choice will depend on how much free gas will be present at the intake for producing condition and on whether there is a packer preventing gas production up the annulus.

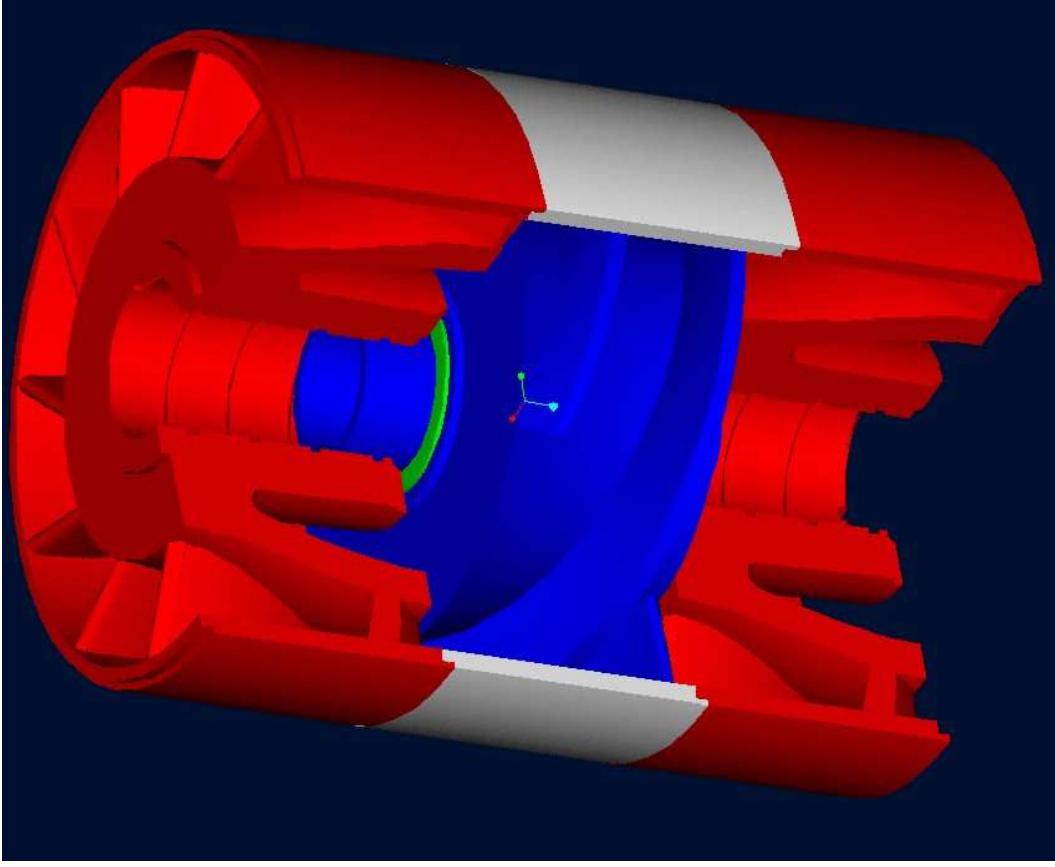


Poseidon Pump

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Poseidon Multiphase Pump Axial Flow Hydraulic Design

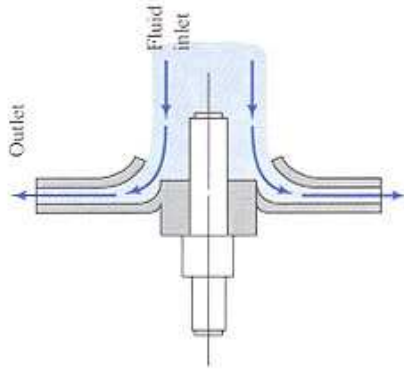


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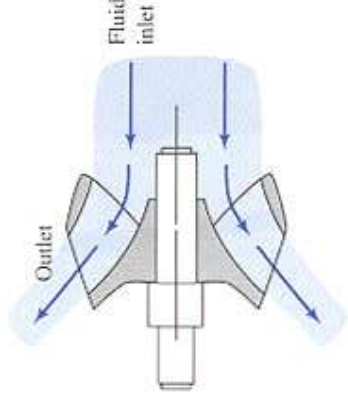
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Poseidon™ Multiphase Pump

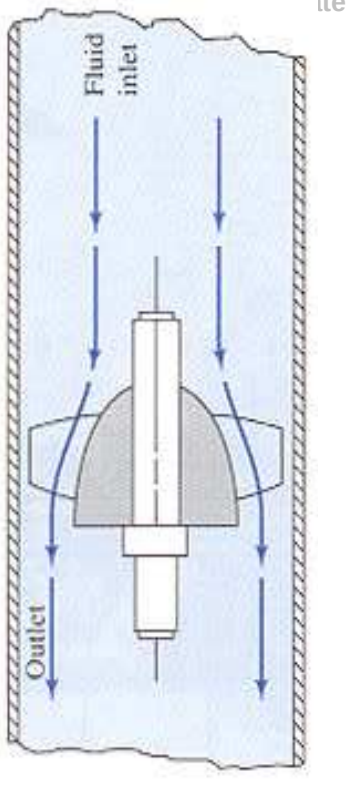
Axial Flow Hydraulic Design



Radial Flow
10% Free Gas



Mixed Flow
20 – 25 % Free
Gas



Axial Flow
70% Free Gas

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Poseidon™ Multiphase Pump

Hydraulic Design

The special stage design allow fluid and gas axial flow which significantly reduces the possibility to gas lock.

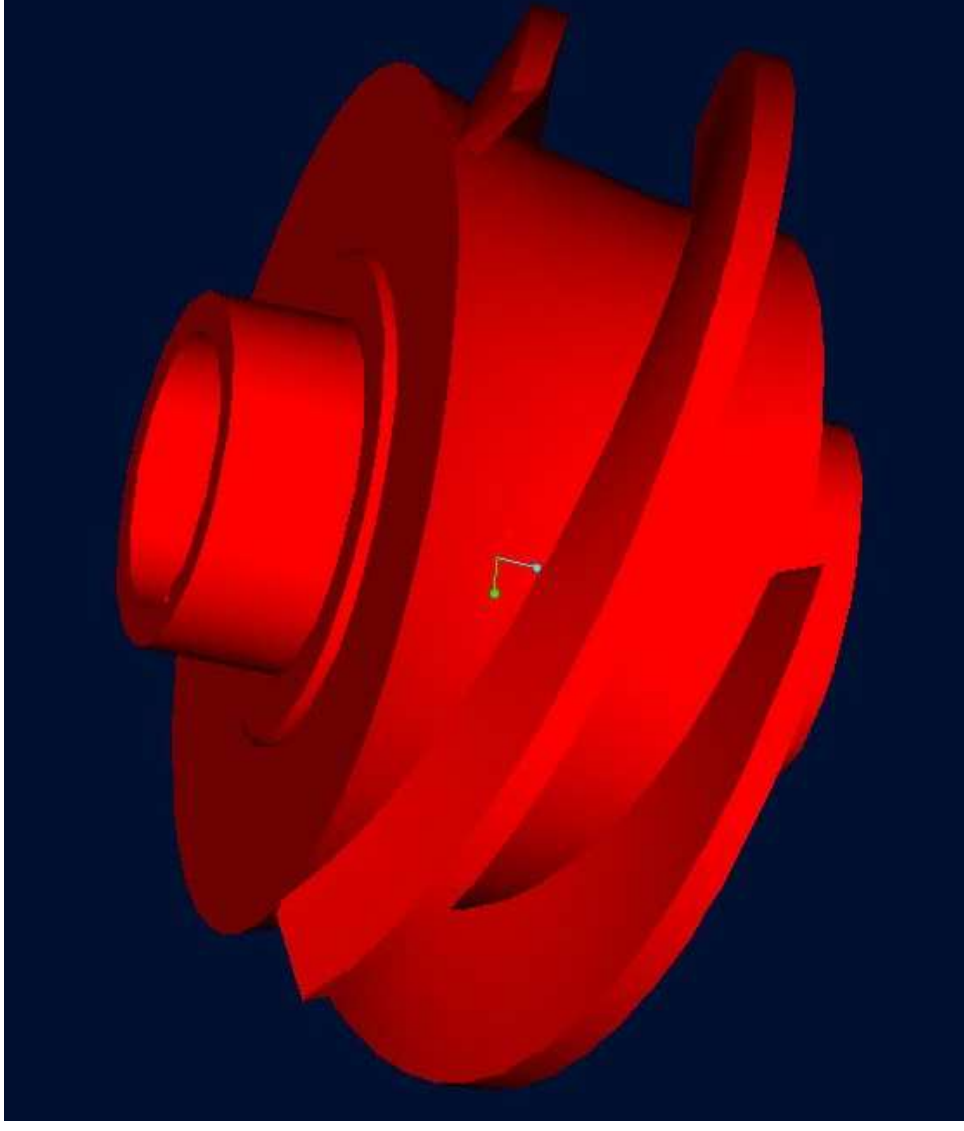
- Homogenizes the fluid to ensure good mixture between gas and liquid.
- Breaks large gas bubbles.
- Increases the mixture pressure to reduce the gas volume.
- Primes the main production pump and pushes the gas/liquid mixture through the production pump stages.

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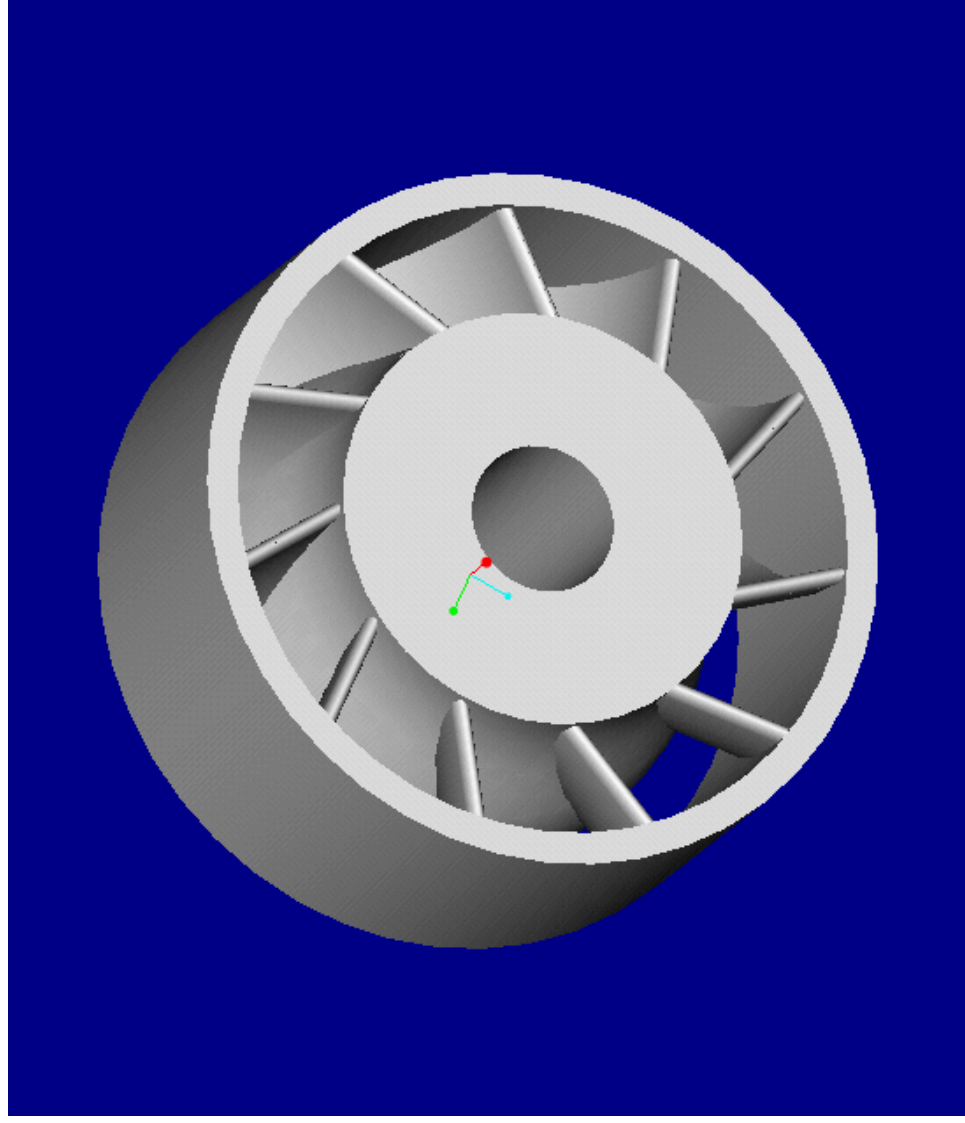
Poseidon Impeller

The special stage design allow fluid and gas axial flow which significantly reduces the possibility to gas lock



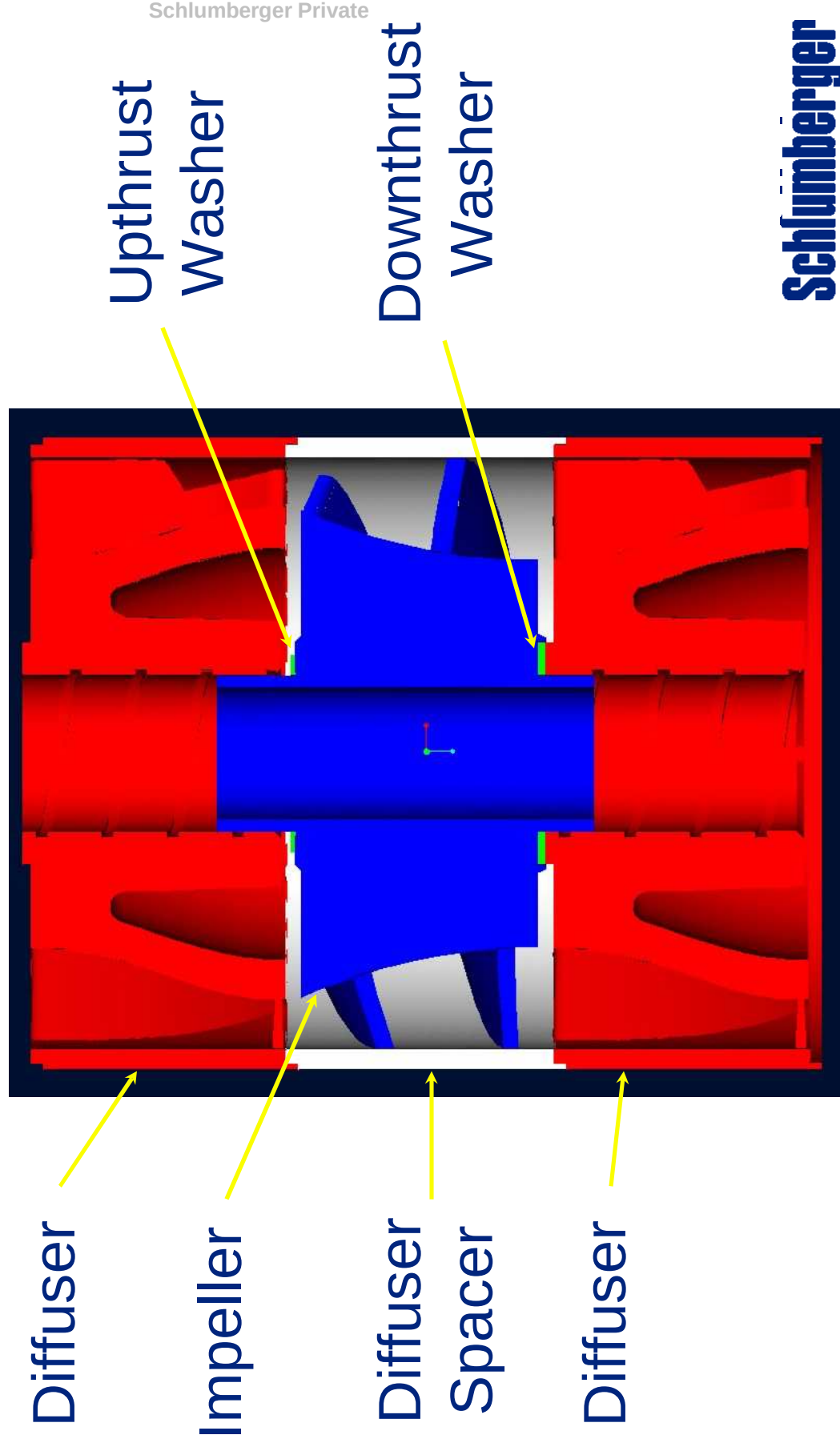
Poseidon Diffuser

The fluid will still flow axially through the diffuser which allows easy gas flow path.



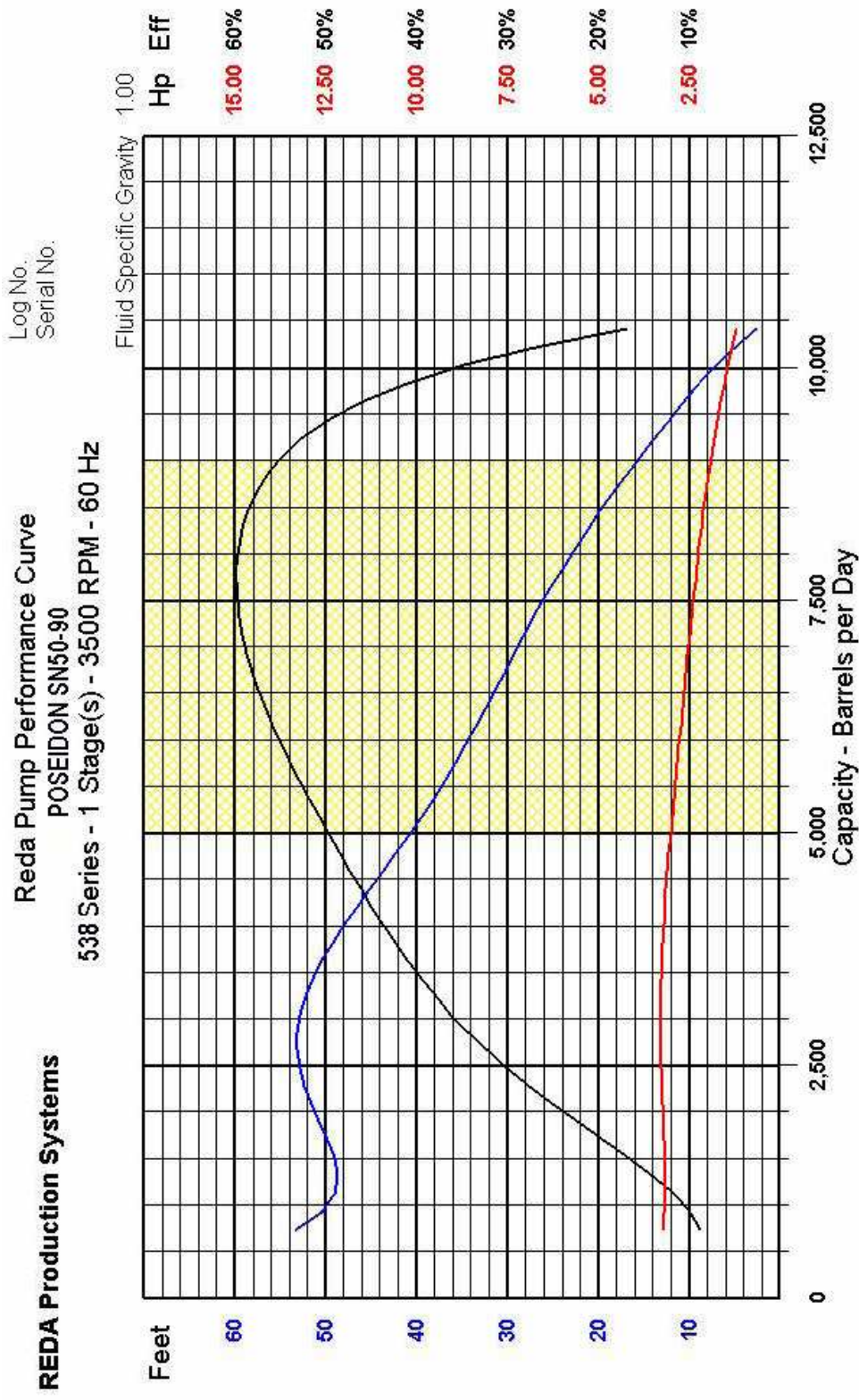
Poseidon Multiphase Pump

Mechanical Design



Poseidon™ Multiphase Pump

Performanced Curve



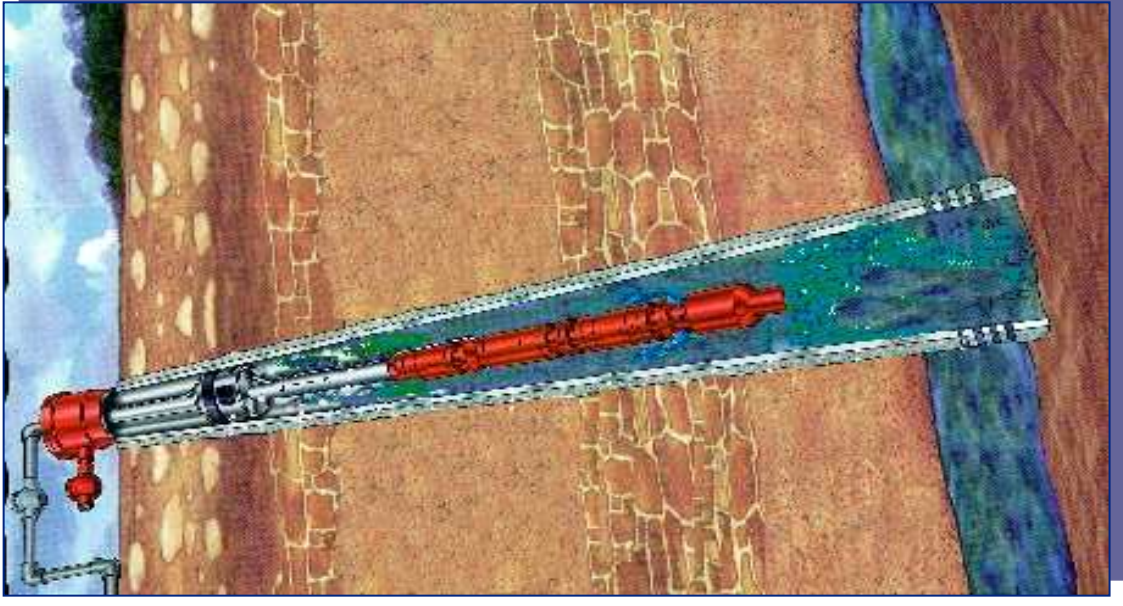
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AGH & Poseidon Benefits

Produce wells with Non vented Packers Above ESPs

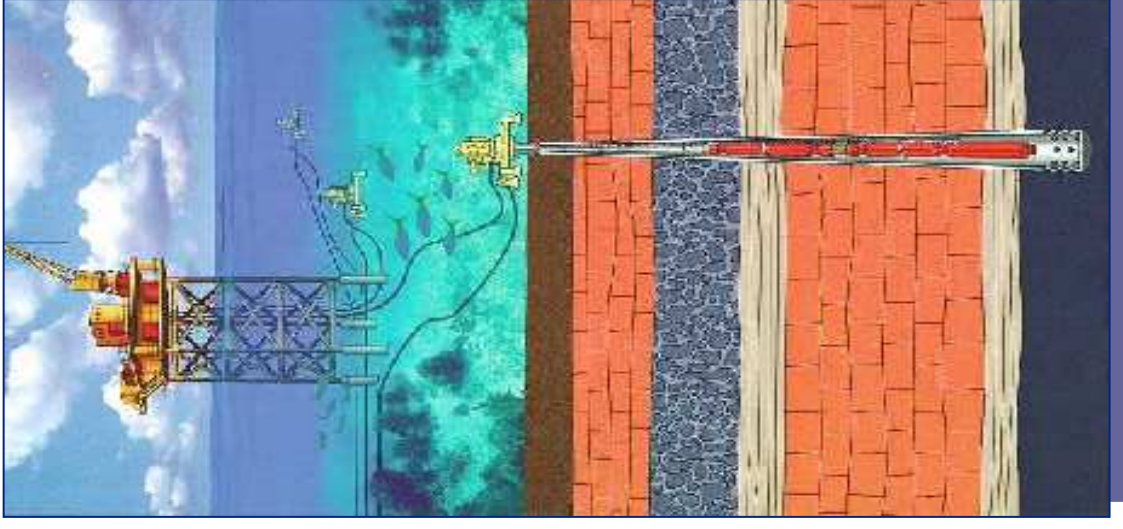
Packers are often installed above pumping units in sensitive areas where subsurface safety valves are required.



AGH & Poseidon Benefits

Subsea Wells

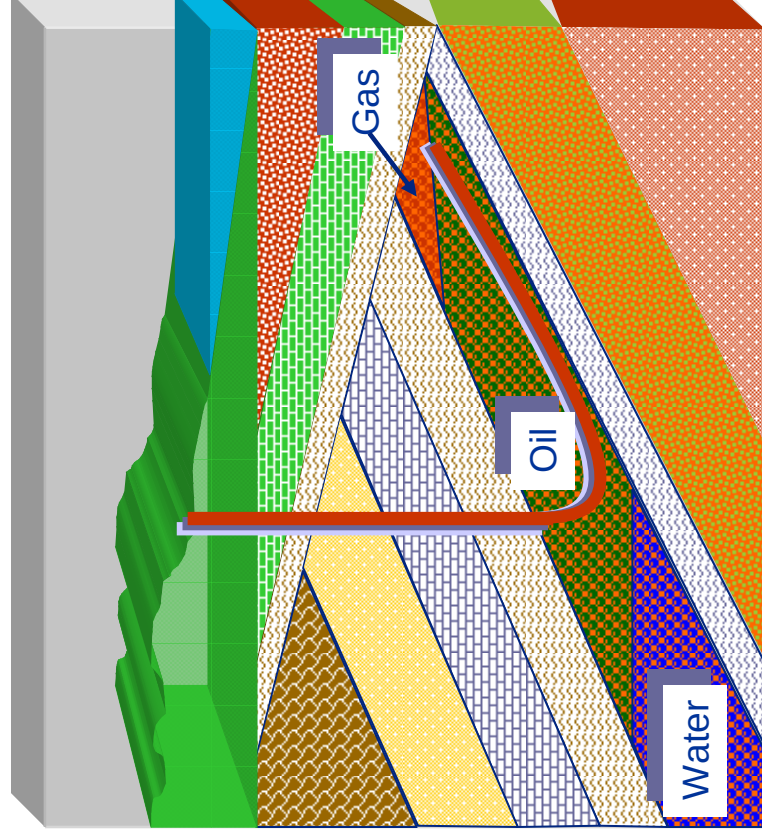
Packers should always be installed above the pumping unit in subsea wells. A Poseidon allows the gas to be produced through the pump into a common production tubing and umbilical. This reduces installation complexity and expense.



AGH & Poseidon Benefits

In horizontal completions the wellbore actually runs parallel with the producing formation. If the wellbore is running updip, gas accumulation in the casing can cause many production problems including slug flow which can lock a pump. A Poseidon can eliminate these problems.

Formation Profile



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AGH & Poseidon Benefits

- Produce wells which could have never been produced by ESP
- Allow all gas to go through the production string to protect the casing
- Reduce lifting cost by using the pumped gas to lift oil in the tubing
- Reduce operating expense
- Improved equipment reliability
- Decreased power requirements in some applications

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